



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.AT.9

Write and Represent Inequalities

Lesson 8-4 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can write inequalities to represent real-world situations.

Language Objective: I can explain my inequality using the words inequality, greater than, less than, and at least / at most.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Write Inequalities

Inequality

Greater than

Less than

At least / At most

Variable

No more than



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Use $<$, $>$, $\leq$ or $\geq$ when the answer is a of values rather than one.

2

A math sentence comparing values with $<$, $>$, $\leq$, or $\geq$ is an

.

3

The symbol $>$ that means one value is larger than another means

.

4

The symbol $<$ that means one value is smaller than another means

.

5 'At least' means greater than or equal to, and 'at most' means

than or equal to.

6 A letter that stands for an unknown number is a .

7 The phrase that means less than or equal to is "no more than," written with the

symbol.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What speeds are possible for the getaway car? How did the phrase 'at most' help you write the inequality?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Write Inequalities** — Represent a range of possible values with $<$, $>$, $\leq$, or $\geq$.

Escribir desigualdades — Representar un rango de valores posibles con $<$, $>$, $\leq$ o $\geq$.

- ☐ **Inequality** — A math sentence that compares two sides with $<$, $>$, $\leq$, or $\geq$.

Desigualdad — Una oración matemática que compara dos lados con $<$, $>$, $\leq$ o $\geq$.

- ☐ **Greater than** — The $>$ sign, showing one number is bigger.

Mayor que — El signo $>$, que muestra que un número es mayor.

- ☐ **Less than** — The $<$ sign, showing one number is smaller.

Menor que — El signo $<$, que muestra que un número es menor.

- ☐ **At least / At most** — 'At least' means $\geq$. 'At most' means $\leq$.

Al menos / A lo más — 'Al menos' significa $\geq$. 'Como máximo' significa $\leq$.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

A speed of 65 mph is ____, a speed of 70 mph is ____, because 'at most 65' means the speed must be ____ 65.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The age could be 18 or any number greater, so it is not a single value.

Evidence: Students explain that 'at least 18' allows many values (18 and up), so an inequality with $\geq$ captures it while an equation would force one value.

Reasoning: This evidence connects the definition of write inequalities to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- A speed of 65 mph is ____, a speed of 70 mph is ____, because 'at most 65' means the speed must be ____ 65.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Write Inequalities
2. **Notes 2:** Inequality
3. **Notes 3:** Greater than
4. **Notes 4:** Less than
5. **Notes 5:** At least / At most
6. **Notes 6:** Variable
7. **Notes 7:** No more than
8. **Watch for this mistake:** *A common mistake in Write Inequalities is using a strict symbol ($>$ or $<$) for a phrase that actually includes the boundary value. For 'the suspect is at least 18 years old,' students often write $a > 18$ instead of the correct $a \geq 18$ — this wrongly excludes 18 itself, even though 'at least 18' means 18 counts as a match. The same error happens in reverse with 'at most': for 'the getaway bag holds at most 12 items,' students write $n < 12$ instead of $n \leq 12$. Always ask: does the key phrase include the number ('at least,' 'at most,' 'no more than,' 'no fewer than' $\rightarrow$ use $\geq$ or $\leq$) or exclude it ('more than,' 'fewer than' $\rightarrow$ use $>$ or $<$)?*
9. **Try It 1:** A. $g \geq 12$ — 'At least 12' includes 12 and any number above it, so $g \geq 12$.
10. **Try It 2:** A. $n < 5$ — 'Fewer than 5' does not include 5, so $n < 5$.